6.3 Producing the process environment to allow the process to take place

The plasma treatment process takes place within the plasma treater enclosure, which is located in the winding section of the machine.

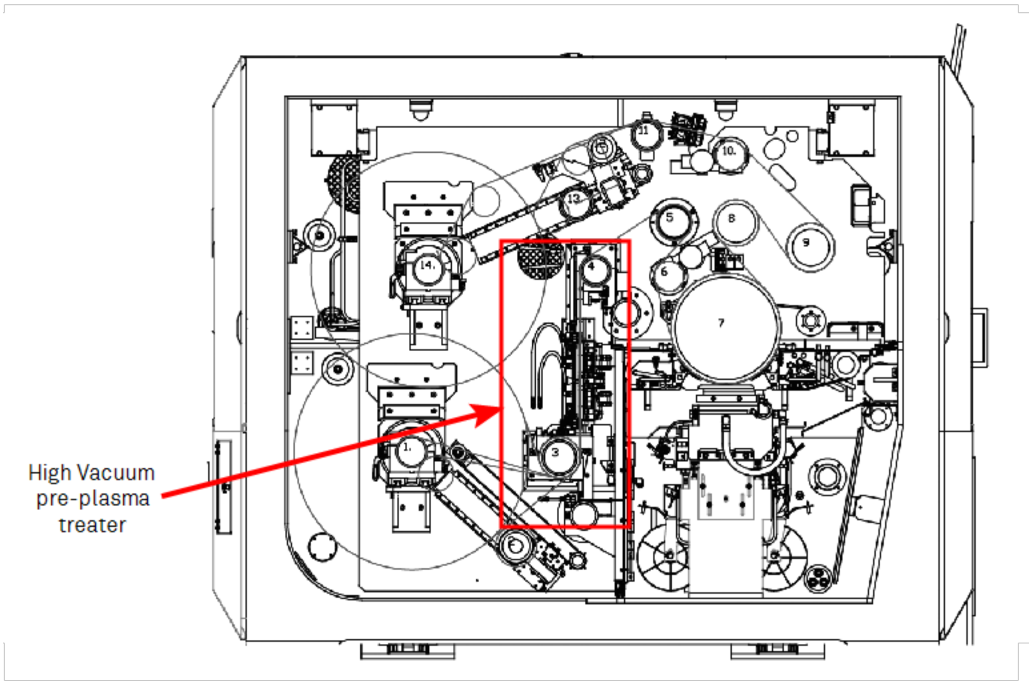

Figure 6.3-1 Cross section of the winding mechanism showing the location of the plasma treaters

The vacuum level in the winding zone of a metallizer will generally below $1e^{-2}$ mbar level, for the plasma process to function. A concentrated process gas is required to be provided, so that the plasma can be formed.

This has to be trapped within the treater assembly.

Pre-treater

The treater assembly is designed to allow even distribution of the process gas within it, and is partly closed to stop the use of excess gas, caused by it being "pumped away" by the vacuum pumping system.

The drawing shows the inner areas of the treater where the gas is supplied.

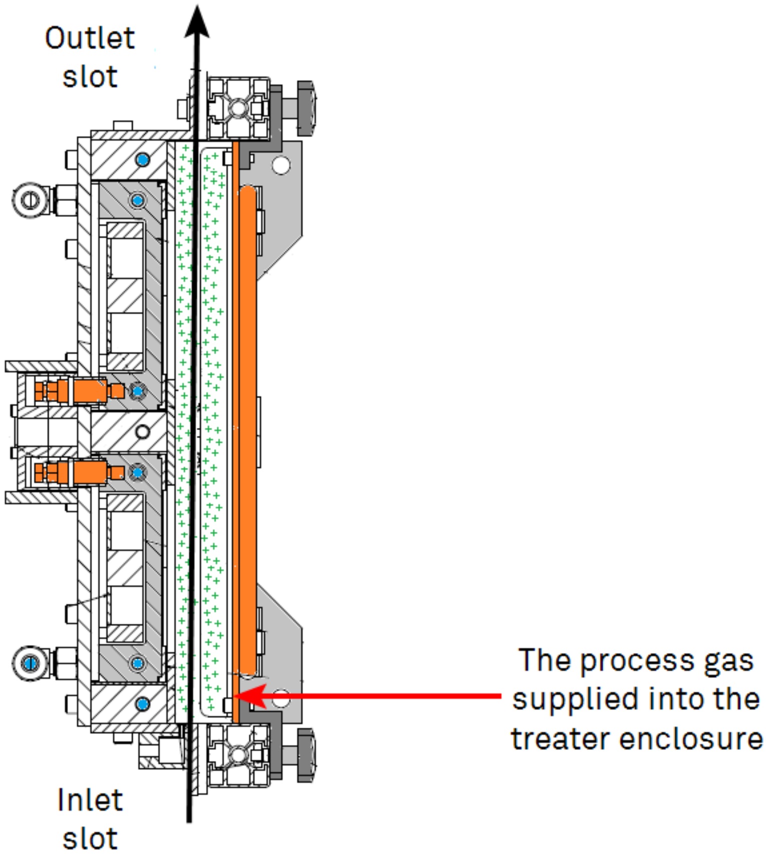

Figure 6.3-2 Plasma treater showing the substrate passing through the treater assembly

Without the sealing front plate sections being installed, the gas can be pumped away easily.

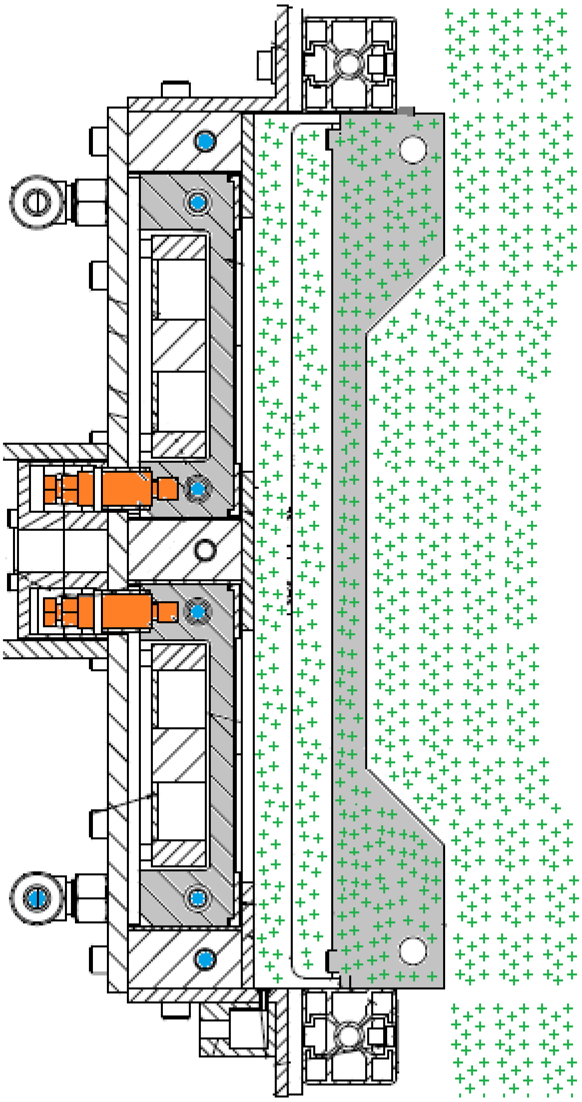

Figure 6.3-3 Example of the treater showing the process gas escaping when the front cover is not installed.

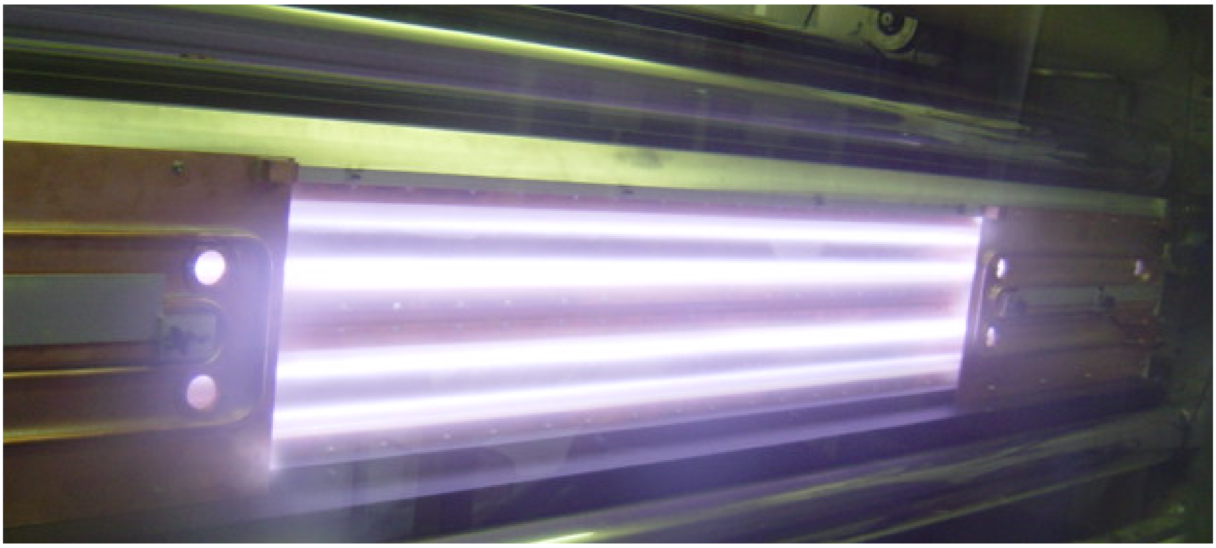

Figure 6.3-4 Pre-plasma treater without the front cover in place, showing how the plasma is formed.

Gas Control System

In order to feed the necessary gas mixtures in specific ratios, a gas control system has been incorporated into the machine.

This consists of one or more mass flow controllers (MFC) and isolating solenoid valves piped to a bottled gas supply.

The MFCs are normally calibrated to the specific gas to which they control. If the MFCs are controlling a different type of gas, different than to what they have been calibrated for, a correction factor is required to adjust the flow level accordingly

Figure 6.3-5 MFC equipment

The process gas can be supplied from industrial gas bottles or other distribution systems, and can be a pre-mixed gas or single types of gas.

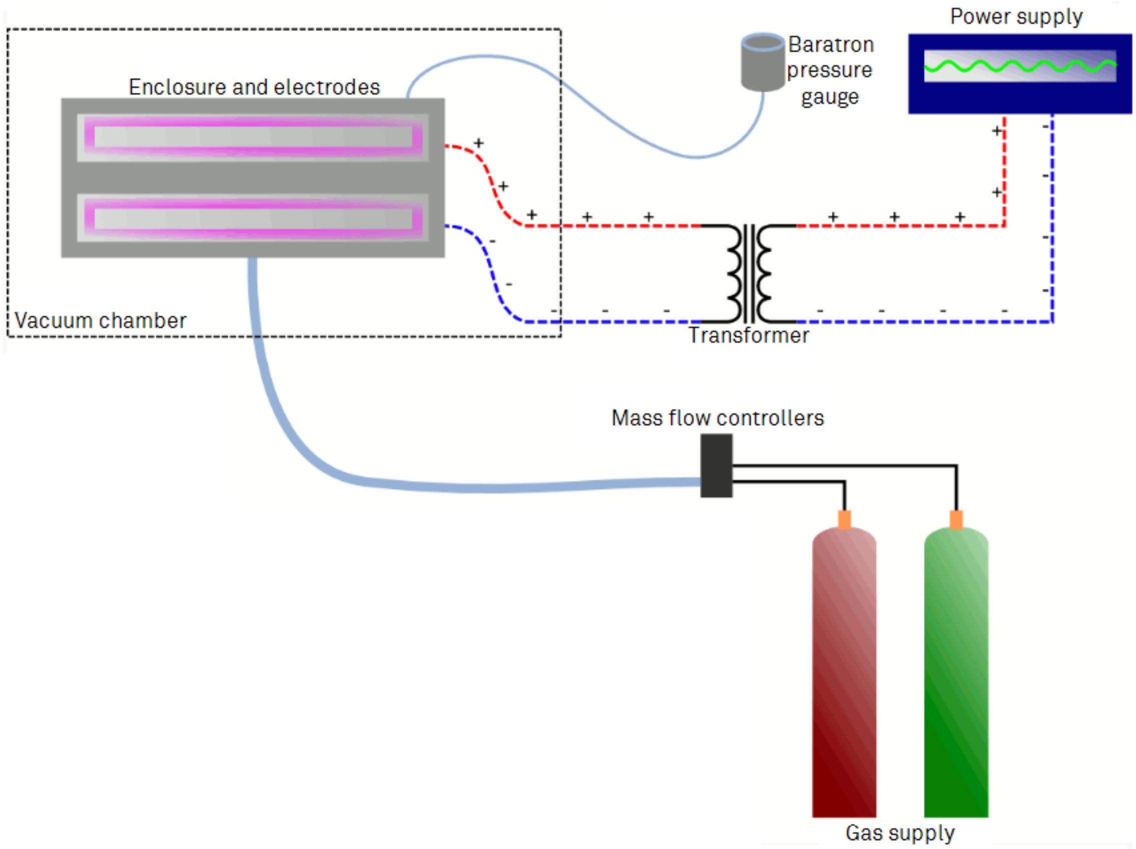

Figure 6.3-6 Process gas supply system

	**DANGER !** **The use and storage of pressurised process gas must comply with national guidelines on handling pressurised systems.**
	**Be aware of potential for any explosion risk that can be caused when using compressed gases.**
	**Be aware of dangers of high concentration of process gases.**

	**Caution !** **Certified gas regulators should only be used and the gas supply to the machine should be limited to 2Bar (28 psi) at the supply regulator.** **Overpressurising the system components can damage them.**

What is the required process pressure in the treater enclosure?

To allow the plasma to be formed inside the plasma treater, a pressure level of between 4E-2 to 8E-2 mbar should be used. With a pressure level outside this range, the treater will not give a constant operation.

How to measure this process pressure?

To pressure level inside the assembly is measured by a Baratron pressure gauge, which displays the pressure level on the DIAGNOSTICS

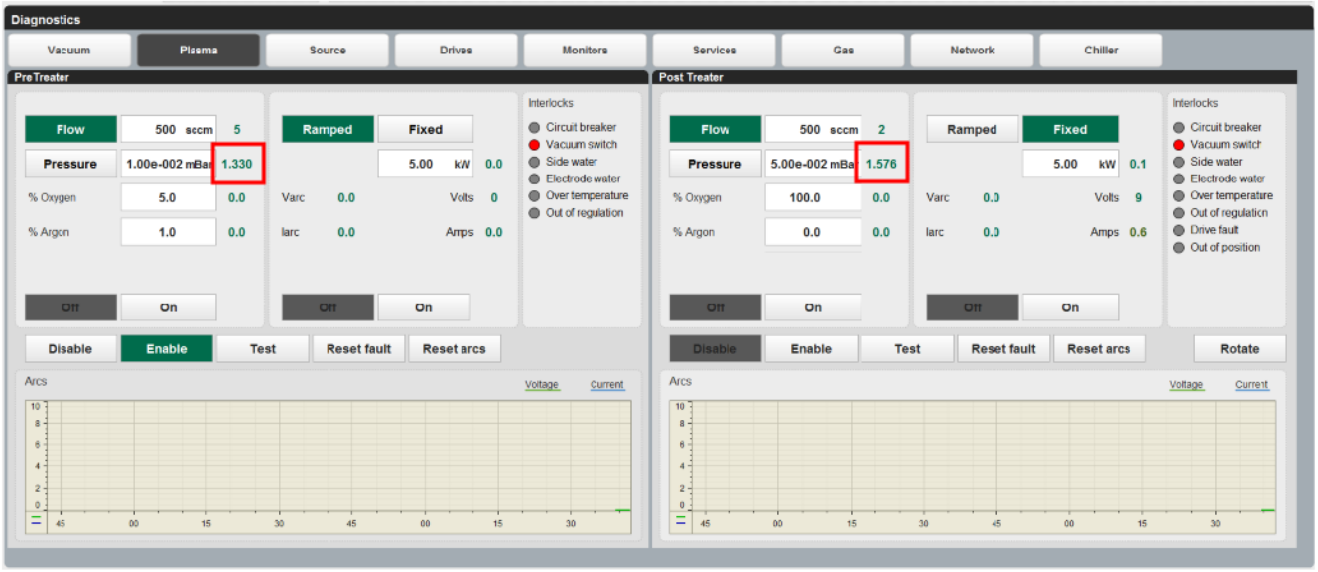

Figure 6.3-7 Gas control section of the DIAGNOSTICS PLASMA HMI screen.

In order to measure the vacuum level of inside the plasma treater zone accurately when using a number of different process gases, a cold cathode Baratron gauge is utilised. Although the gauge is located on the outside of the chamber, an internal flexible tube runs between the gauge connection and the treater assembly.

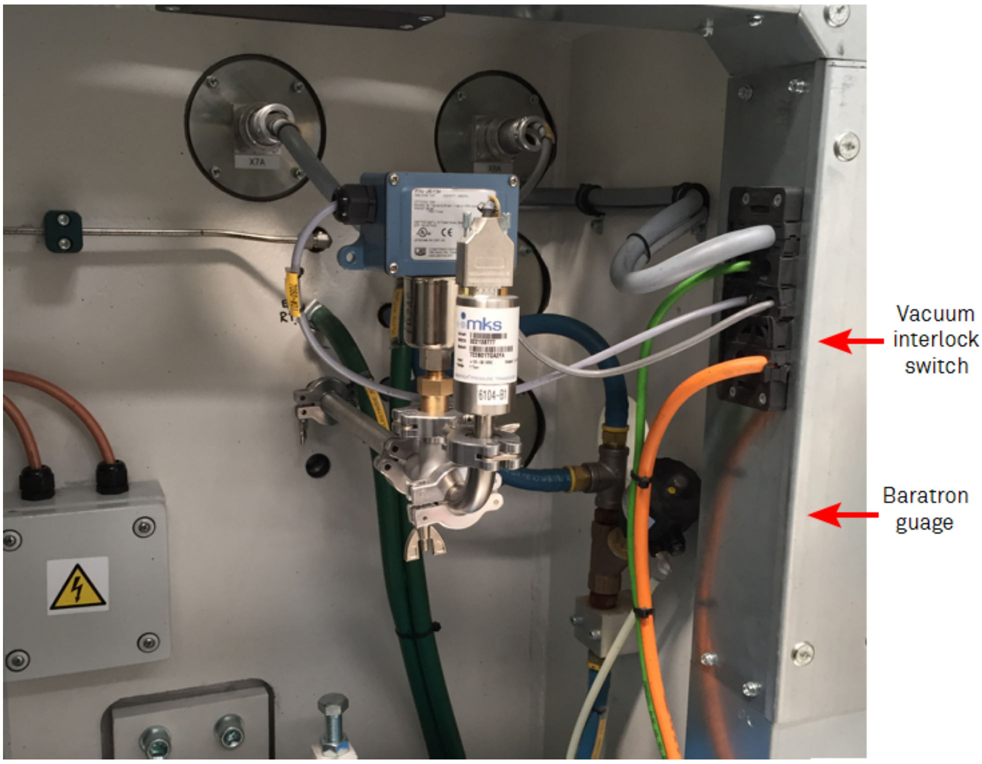

Figure 6.3-8 Location of a Baratron gauge

	Warning !
	Do not adjust or override the safety interlock components.

How is the pressure level controlled?

To allow the plasma to be formed inside the plasma treater, a pressure level of between 4-8e^{-2} mbar should be used, by accurate controlling of a process gas into the treater enclosure.

In order to feed the necessary gas mixtures in specific ratios, a gas control system has been incorporated into the machine.

This consists of one or more mass flow controllers (MFC) and isolating solenoid valves piped to a bottled gas supply.

The MFCs are normally calibrated to the specific gas to which they control. If the MFCs are controlling a different type of gas, different than to what they have been calibrated for, a correction factor is required to adjust the flow level accordingly.

Figure 6.3-9 MFC equipment

Controlling the pressure level within the plasma treater assembly

There are two modes of operation of control that can be used to produce the required process gas environment inside the treater enclosure.

Pressure	Select the operating pressure for the plasma treater for the gas flow to vary to match the desired operating pressure. This is the preferred mode of operation.
Flow	Allows the gas flow to be controlled independently of the pressure in the treater enclosure at a fixed gas flow rate. If the pressure regime inside the enclosure exceeds the 4-8e^{-2} mbar range, it will be expected that high levels, short circuit arcing can occur, or no plasma maybe formed.

The amount of gas that can be set is normally 4000 sccm.

Selecting the Gas Type to be used

The type of process gas to be supplied to the plasma treater is initially selected in the SETUP HMI screen.

The type of common gases that can be selected are-

- Argon
- Nitrogen
- Oxygen
- Mixed gas (80% Argon/ 20% Oxygen)

The selection of the gas type changes the amount of gas flow that passes through the MFC

Note!
The types of gases that can be used are selected based upon the machine specifications

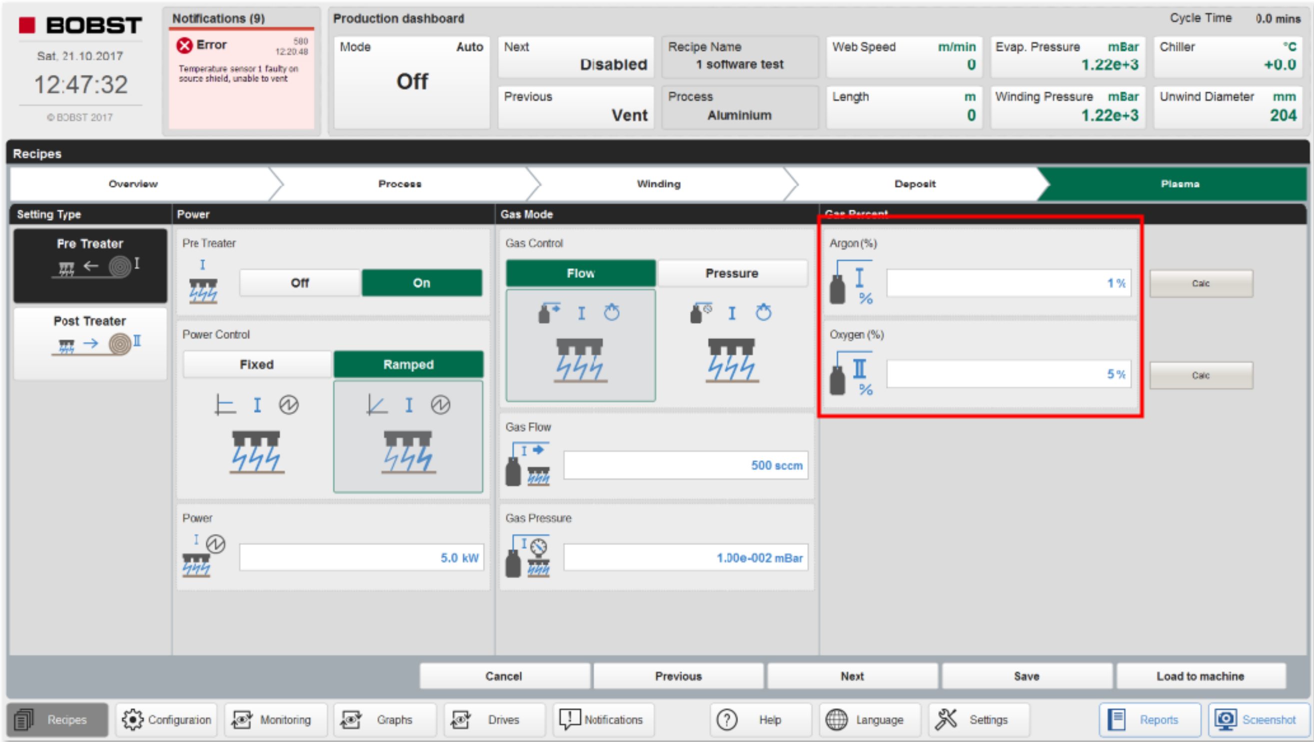

Figure 6.3-10 RECIPES - PLASMA HMI screen.

To allow for a controlled gas mixture to be used, the percentage of the total gas flow can be set by percentage.

Entering a percentage value in one set point parameter and pressing the CALC button will automatically set the other parameter to give a total of 100%.

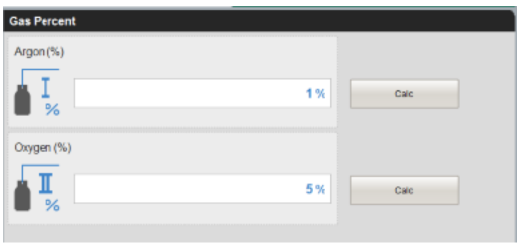

Figure 6.3-11 RECIPIES - PLASMA HMI screen - gas percentages

The control of the gas flow and the mixture of the combined process gas can also be changed while the process is taking place from the CONFIGURATION PLASMA HMI screens

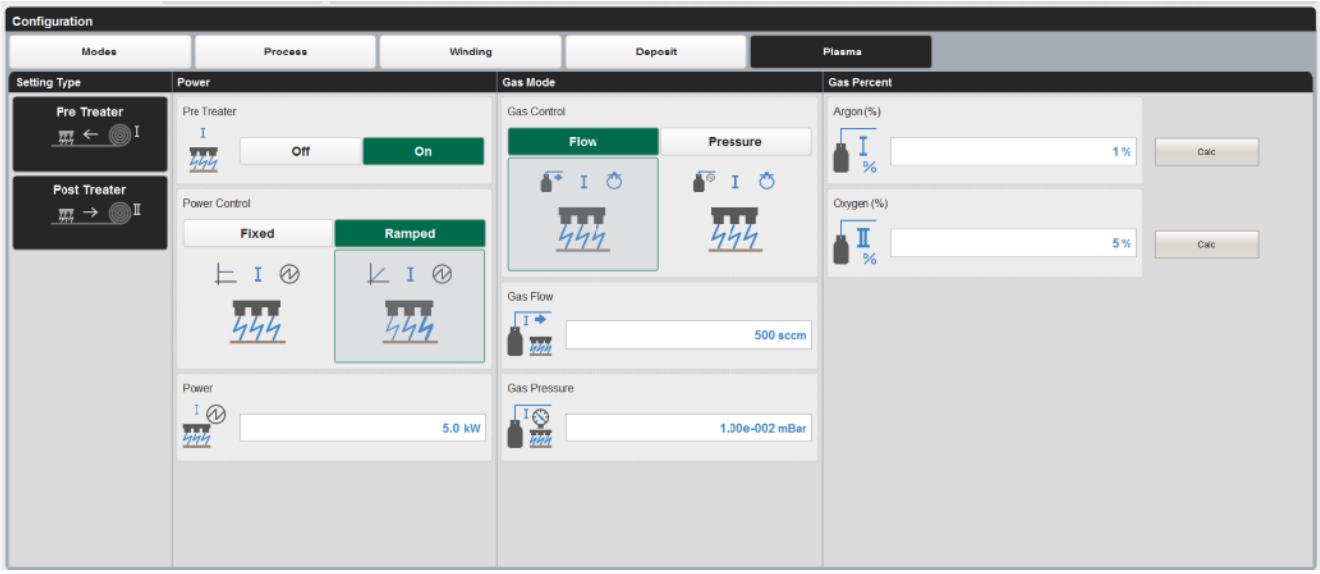

Figure 6.3-12 CONFIGURATION - PLASMA HMI screen

Enabling the plasma gas control

The operation of the gas supply is by selecting the PLASMA TREATERS ON button on the MONITORING – PROCESS HMI screen

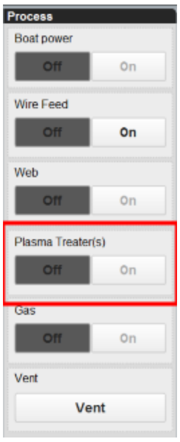

Figure 6.3-13 MONITORING – PROCESS HMI Screen – Enable and reset fault buttons

6.4 Plasma System Power Supply

To create the plasma, an electrical charge has to be placed across the electrodes and the surrounding enclosure.

This machine uses a low frequency AC power supply for this purpose.

Power Supplies Location

The power supply is normally located within an electrical panel on the rear of the winding cart.

The plasma power supplies are all located within the electrical panel designated E0_7.

There is an interface on the master units, but all controls are carried out remotely through the main machine HMI.

Figure 6.4-1 Plasma treater power supply panel, with access door open showing the components

Plasma Treater HMI Interfacing

The control of the plasma treater power supply is from a number of screens on the HMI

To monitor and change the set points, a facility is provided on the MONITORING HMI screen, under the PARAMETERS section.

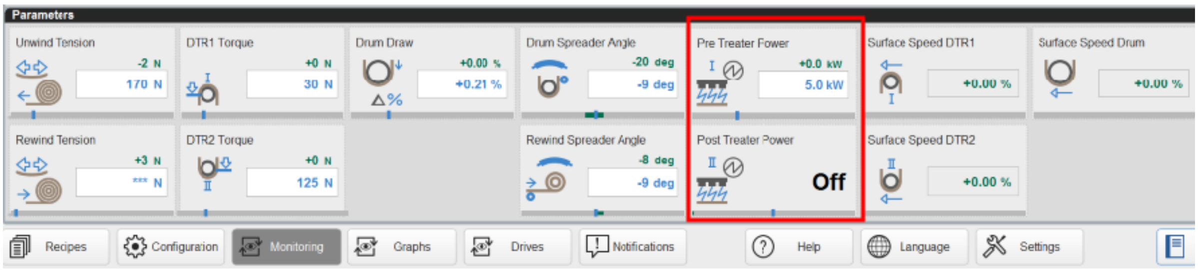

Figure 6.4-2 MONITORING – PARAMETERS HMI screen – Plasma section

The user can monitor the power being applied feedback by the green numbers, while the set point (blue) can be changed.

Note!

The amount of power available to be supplied will depend upon the power supply specified and the process conditions

Plasma Treater HMI Interfacing – Recipes and Configuration Screens

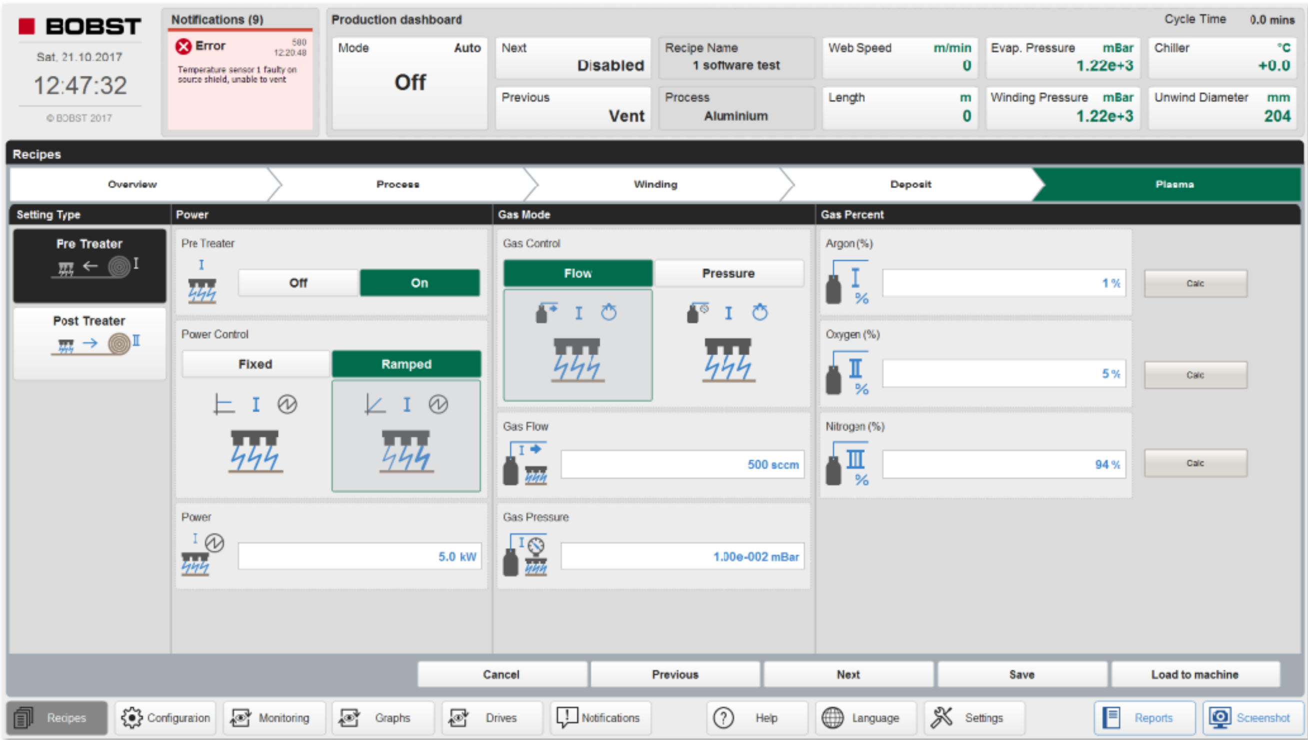

Figure 6.4-3 RECIPES – PARAMETERS HMI screen – Plasma section

This screen allows the user to set the process gas and the power supply control parameters.

The treater can be selected here to be used within the process cycle, by the ON/OFF buttons.

The power supply can be set to apply the full power set point by selecting FIXED, or can allow the set point to be ramped up from zero to the set point, by selecting the RAMPED mode of operation.

The power set point can be entered in the box, by touching it and a pop up numeric box will allow the value to be changed.

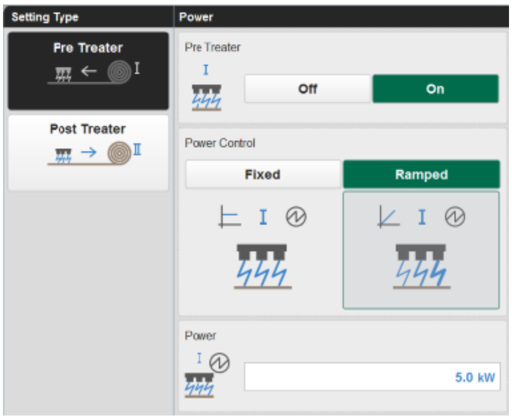

Figure 6.4-4 RECIPES – PARAMETERS HMI screen

These parameters are set when the recipe is loaded to the machine.

Once loaded and the process has been started, a quick access to a limited selection can be through the CONFIGURATION HMI screens.

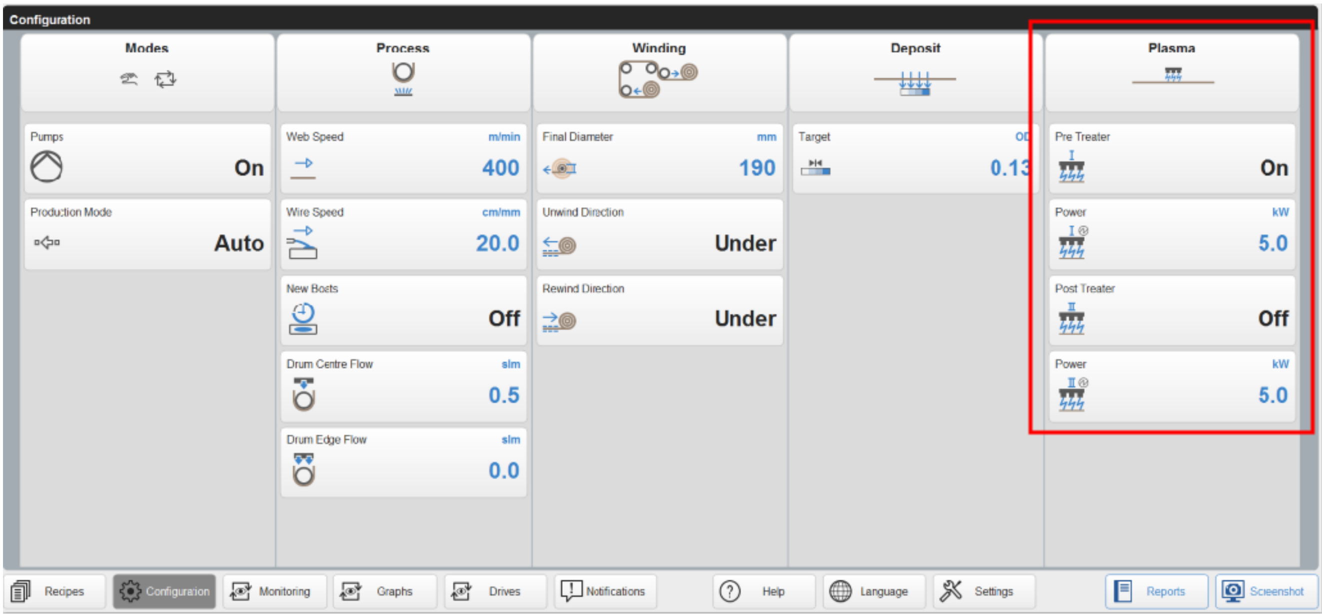

Figure 6.4-5 CONFIGURATION - PLASMA - OVERVIEW HMI screen

There is further screens under the CONFIGURATION section that allow changes to the gas flow, power settings, etc.

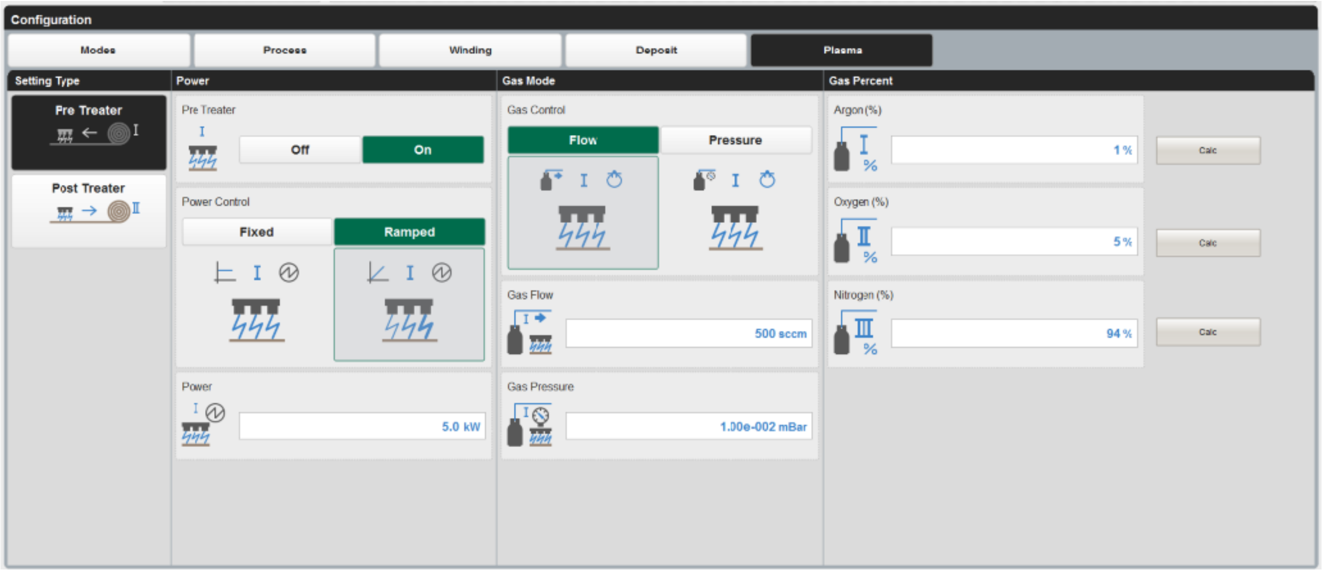

Figure 6.4-6 CONFIGURATION - PLASMA - HMI screen

Plasma Treater Diagnostics

Diagnostics of the system can be monitored through the SETTINGS- DIAGNOSTICS- PLASMA HMI screen.

Figure 6.4-7 DIAGNOSTICS - PLASMA - HMI screen

This screen gives the user the ability to understand how the plasma treater(s) are performing in detail.

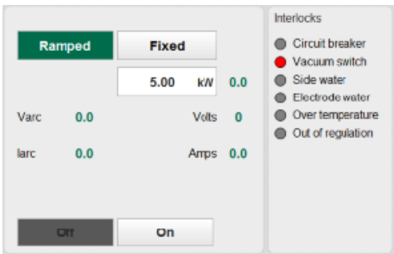

Figure 6.4-8 DIAGNOSTIC - PLASMA - HMI screen

Some of the operational parameters available on this screen have already been described in this section of the manual.

The conditions are generated from the power supply and reference should be made to the relevant vendor manual.

Alarm	Reason
Circuit Breaker	This alarm will be shown if the circuit breaker trips under load. It is located in panel E0_7_0
Vacuum Switch	This is an indicator that the chamber is under vacuum and that it is safe to operate the plasma treater
Side Water	This alarm will be indicated if the water flow to the plasma treater side falls below 5 litres/min.
Electrode Water	This alarm will be indicated if the water flow to the plasma treater electrode falls below 5 litres/min.
Over temperature	This alarm will be indicated if the plasma treater power supply overheats. The power supplies on this machine are water cooled, so checks should be made to find out if water is flowing.
Out of Regulation	This alarm is indicated when the actual power to the plasma treater does not match the set-point required. There may be a number of process related aspects that can cause this issue.

Table 6.4-1 List of Interlocks – AE Plasma Treater

When a short circuit (arc) occurs it is counted and displayed either as a voltage arc or a current arc on the display below.

It is scaled number of arcs on the “Y axis” and time on the “X” axis over 1 minute.

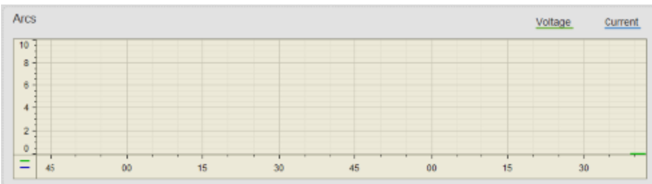

Figure 6.4-9 DIAGNOSTIC - PLASMA - HMI screen – Indication of arc instances

Button	Description of purpose
Disable	This button is provided to disable the plasma treatment function.
Enable	This button is provided to enable the plasma treatment function.
Test	There is a requirement when initially powering up the plasma treater to allow its operation to be checked out. Pressing the ‘test button’ allows the user to power up the plasma treater without any web in the winding mechanism, to allow clean-up of the target plates. There is an interlock indication panel to allow the user to ensure that all necessary interlocks are made. If this mode is to be used, remove the substrate from the machine, unless machine closed and running substrate. Selecting the test button will allow the operation of the plasma treater. All the interlocks have to be present for the plasma treater to be powered up.
Reset fault	This button is used to reset the gas control circuit in the case that any faults occur.
Reset arcs	This button is used to reset the current and the voltage arc counters.

Table 6.4-2 List of Interface buttons – common to planar and rotatable systems

	Note!
	If the ‘Plasma Test button’ is left selected, tension cannot be applied to the substrate.

6.5 Plasma Process Setup - HMI

The parameters to allow the pre and post treaters to be used as part of the overall coating process are inputted in the HMI SETUP screen. These are accessed from the top screen selection tabs on the HMI.

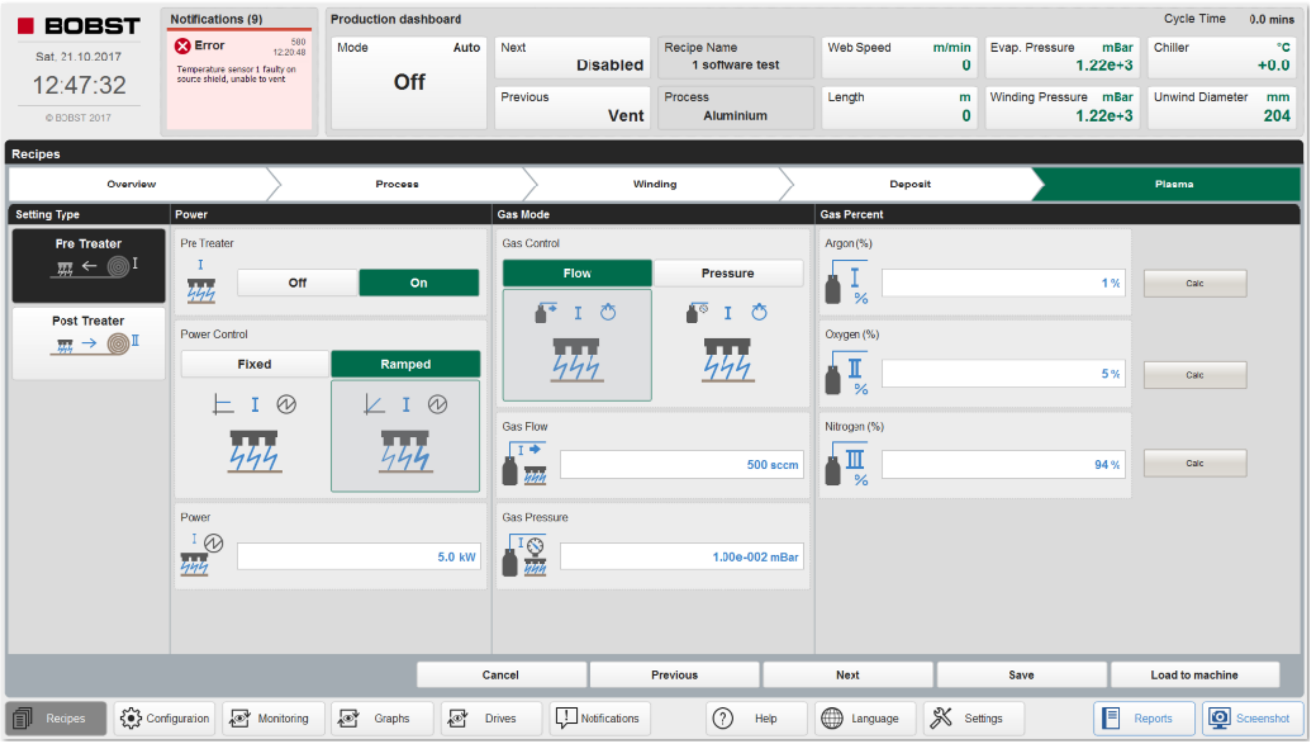

Figure 6.5-1 RECIPES – PARAMETERS HMI screen – Plasma section

Power Set-point Control:

The power supply operation power set-point level can be set by pressing the blue box, to bring up the numeric keyboard. The value that can be entered will be limited to the plasma power supply output rating.

Controlling the pressure level within the plasma treater assembly

There are two modes of operation of control that can be used to produce the required process gas environment inside the treater enclosure.

Select the operating pressure for the plasma treater for the gas flow to vary to match the desired operating pressure. This is the preferred mode of operation.

Allows the gas flow to be controlled independently of the pressure in the treater enclosure at a fixed gas flow rate. If the pressure regime inside the enclosure exceeds the 4-8e-2 mbar range, it will be expected that high levels, short circuit arcing can occur, or no plasma maybe formed.

The amount of gas that can be set is normally 4000 sccm.

Selecting the Gas Type to be used

The type of process gases to be supplied to the plasma treater is also initially selected in the RECIPES PLASMA HMI screen.

The type of common gases that can be selected are-

- Argon
- Nitrogen
- Oxygen
- Mixed Gas (Example - 80% Argon/20% Oxygen)

The selection of the gas type changes the calibration value for the gas flow that passes through the mass flow controller(s) (MFC).

	Note !
	The types of gases that can be used are selected based upon the machine specifications

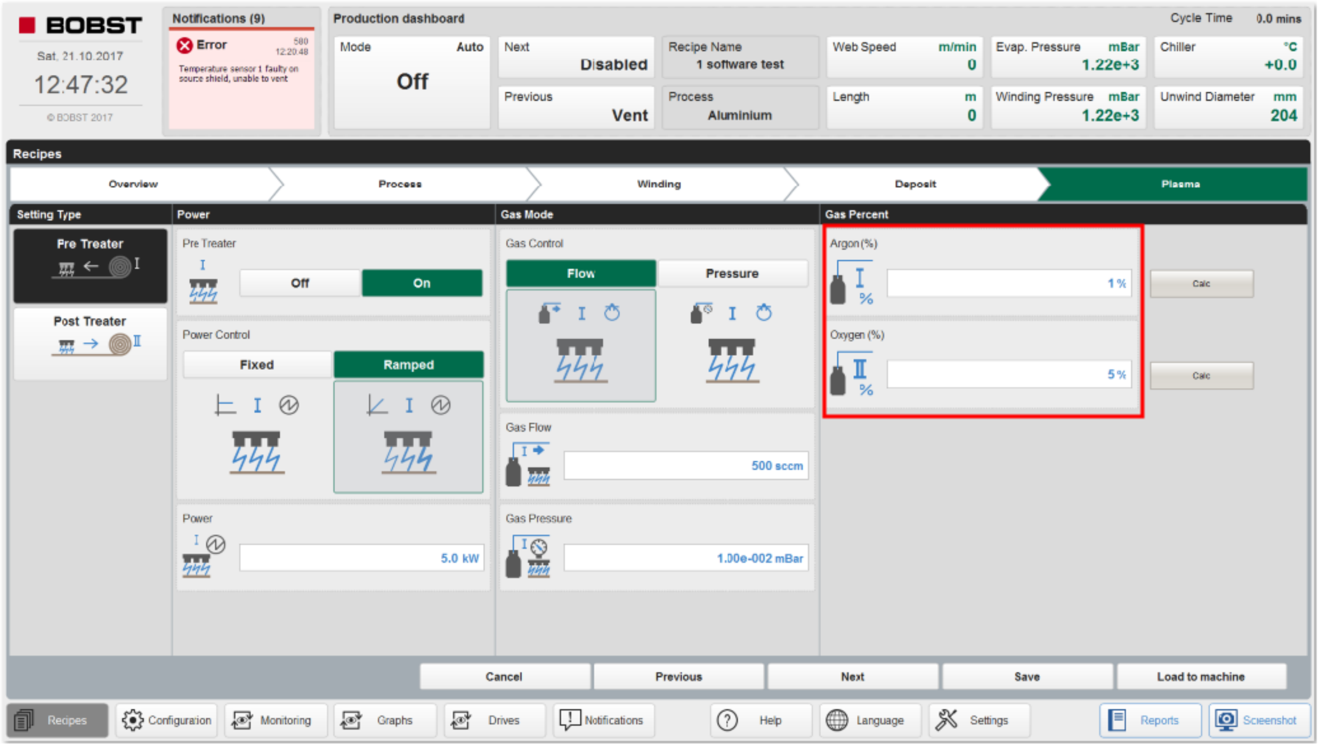

Figure 6.5-2 RECIPIES - PLASMA HMI screen

To allow for a controlled gas mixture to be used, the percentage of the total gas flow can be set by percentage.

Entering a percentage value in one set point parameter and pressing the CALC button will automatically set the other parameter to give a total of 100%.

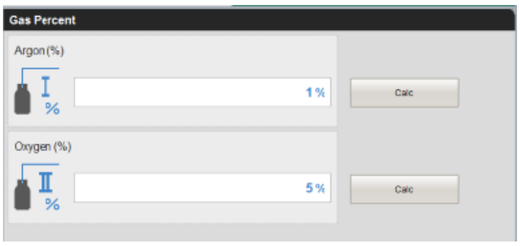

Figure 6.5-3 RECIPIES - PLASMA HMI screen - gas percentages

	Note !
	Depending upon the process gas mixture used and the type of film being processed, the full power rating of the system may not be available due to restrictions in the "electrical" inductance of the system. **If the actual winding zone pressure is not below the plasma pressure set-point, no gas will flow, and no plasma can be generated.**

	Caution !
	Operating at lower or higher pressures outside the recommended levels will potentially lead to excessive erosion by sputtering, at low pressures, or excessive arcing at higher pressures.

6.6 Water Cooling and HMI Interfacing

As there is energy being created within the plasma treater assembly, the system has to cooled by water, to stop physical damage by overheating.

	**Caution !**
	Take care not to damage this unit by trying to operate without the correct water-cooling

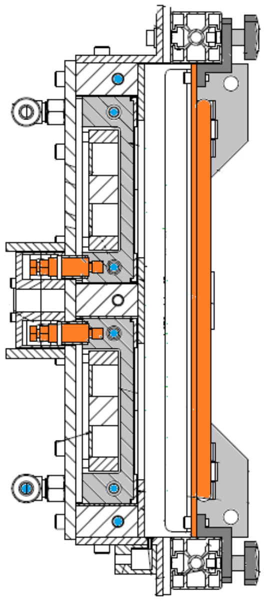

Figure 6.6-1 Cross section of internal water-cooling, as indicated by the blue areas plus the cooling on the doors

Water Supply Isolation

To stop water condensation being formed on the plasma treater, the water is isolated when the winding cart is withdrawn from the chamber. This is carried out via a valve in the supply line.

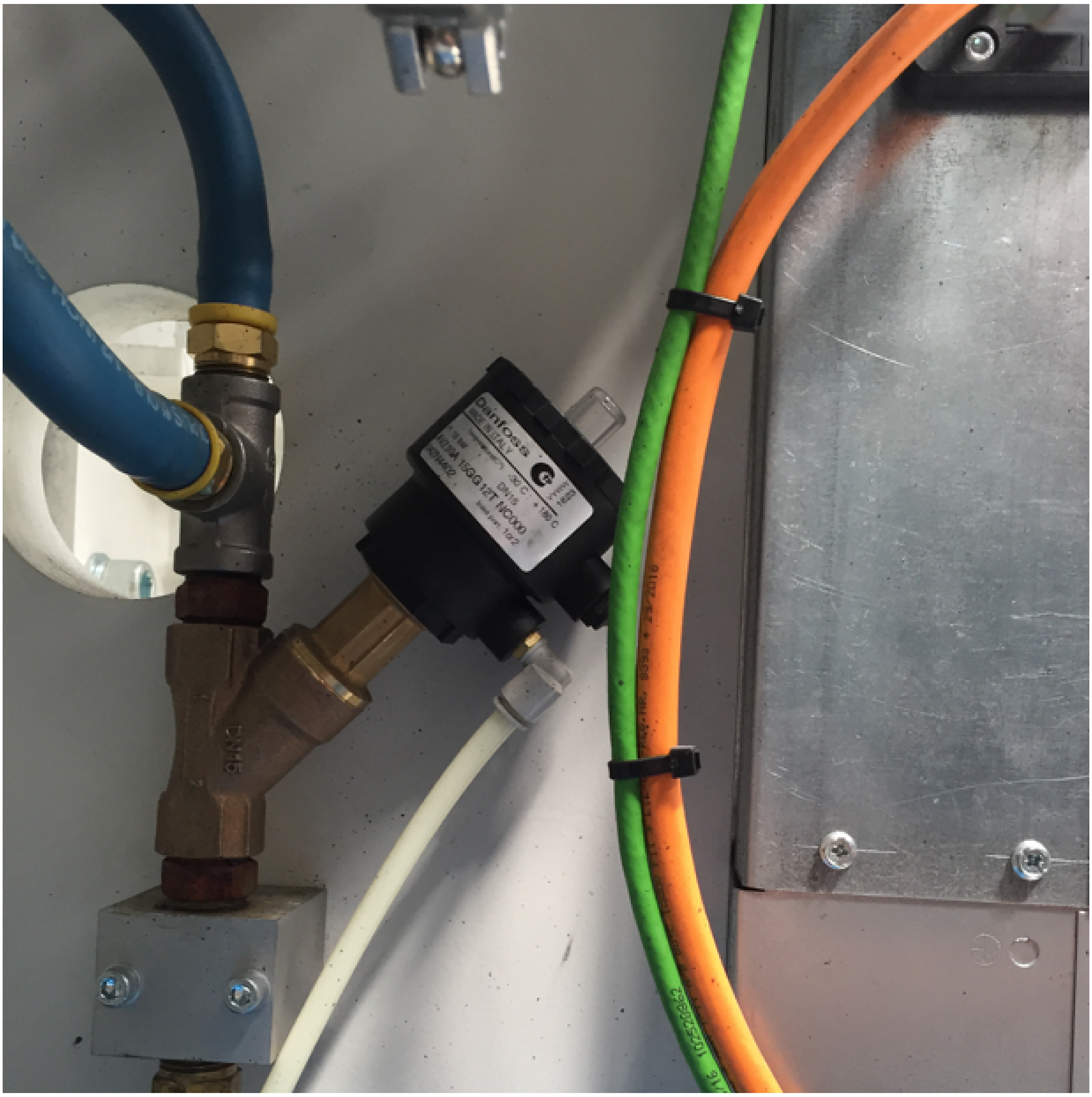

Figure 6.6-2 Water isolation valve

Use the water supplies section of the services diagnostics screen to operate the isolation valve when the winding cart is fully retracted from the chamber:

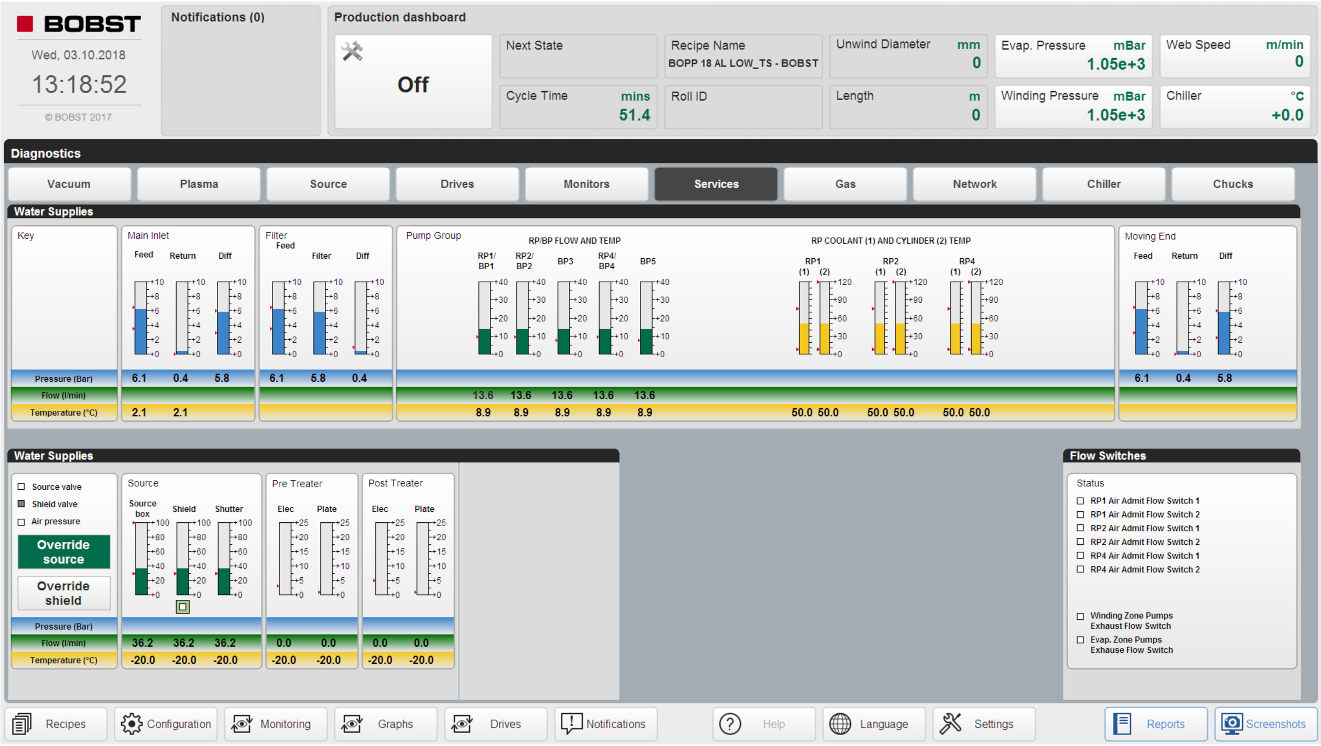

Figure 6.6-3 Services Diagnostics screen

Flow and temperature measuring devices

Combined flow and temperature transducers are supplied for measuring the return flows from the treater(s)

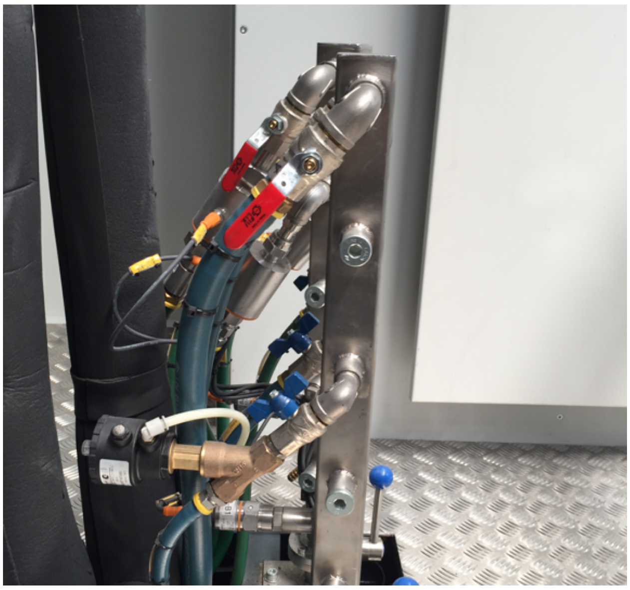

Figure 6.6-4 Water supply manifold indicating the water flow monitors

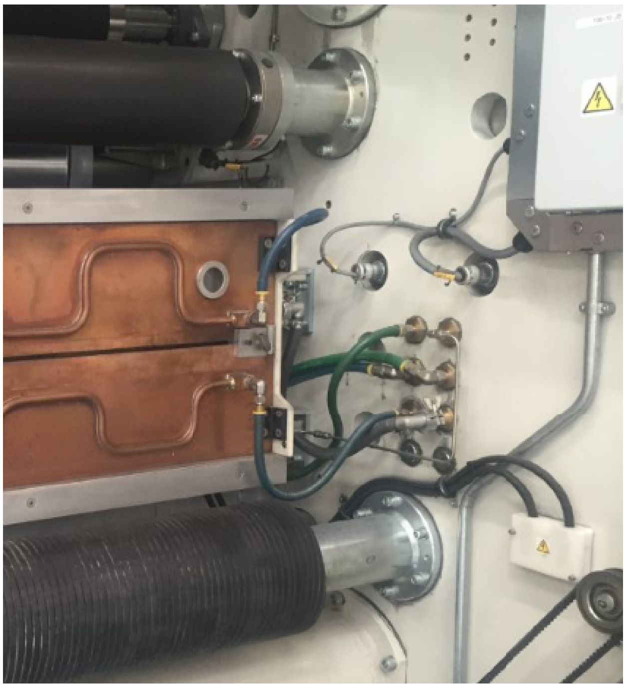

Figure 6.6-5 The inter water lines at the plasma treater

Caution !

Do not shorten the length of the flexible water hoses or replace with any alternative, as it will affect the performance of the system

Power Supply Cooling

The plasma power supplies are water-cooled.

These supplies are isolated when the power supply is switched off.

The power supplies have internal temperature monitoring and if there is a problem with either, an alarm will be shown on the plasma diagnostic HMI screen.

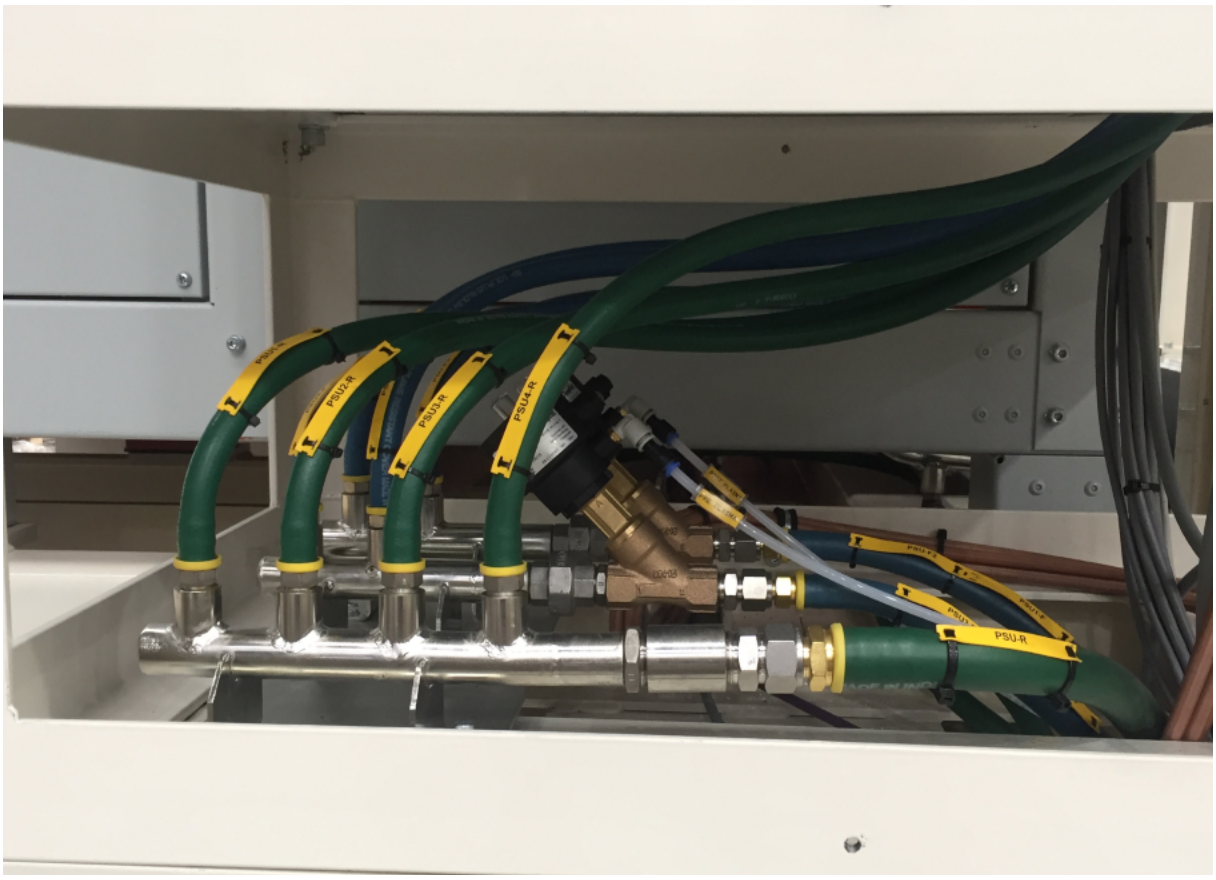

Figure 6.6-6 Water isolation valves that are located below the plasma power supply enclosure

HMI Interfacing and Interlocking

The flow and temperature measurements are communicated back to the machine control systems

If there is a problem with either, an alarm will be shown on the plasma diagnostic HMI screen.

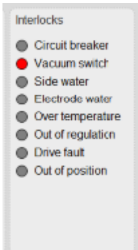

Figure 6.6-7 DIAGNOSTICS - PLASMA HMI screen.

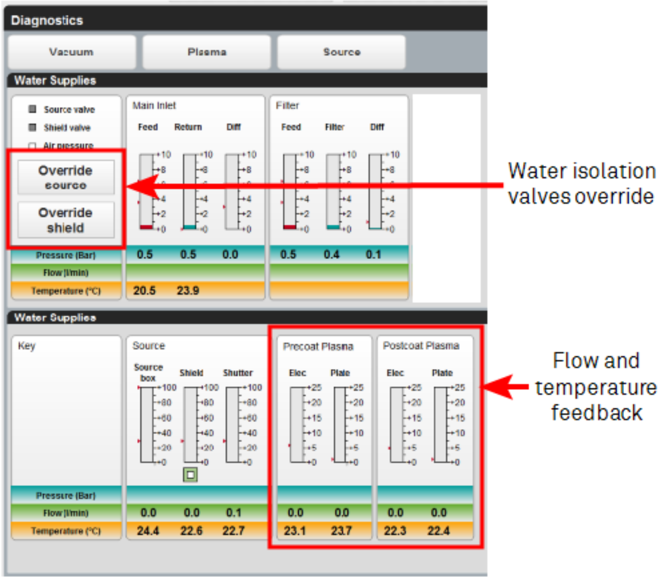

Figure 6.6-8 DIAGNOSTIC - WATER SERVICES HMI screen indicating sensor readings.

Water Isolation to Internal Cooled Circuits

To assist in the reduction of condensation on the machine, the water to the coating shield, plasma system and evaporation source is isolated once the machine has been opened fully.

	Note !
	**At the maintenance log level, these valves can be activated by touching the icon on the screen, when the winding cart is fully retracted.** **If the chamber is closed and the rotary pump RP1 is not running, no water will be circulated.**

6.7 Setting the Anti-Backside Treatment Plasma Shields

This chapter describes how to set the anti-backside treatment plasma shields.

When using the plasma treater, there is a requirement only to treat the side of the substrate that the aluminium is to be deposited on.

As there is no control of where the process gas is distributed within the treater enclosure, there is a possibility that the plasma can be also be present on the rear side of the substrate. This can later result into a blocking problem, where layers of the substrate stick together, which can cause permanent damage.

In order to prevent this "backside" treatment of the substrate, a pair of backside treater or plasma shields has been incorporated into the design. These can be accessed from the front of the treater, while the substrate is present.

Figure 6.7-1 Photograph showing the vertical anti-back side treatment shield on the door of the treater

The adjustment blocks for the backside shields, should be moved to be behind the substrate edges for about 15mm, when the substrate has been centred in the machine.

Release the clamping mechanism and remove the focus shield

Figure 6.7-2 Photograph showing user loosening the shield clamping plate

Figure 6.7-3 Photograph showing user removing the focusing shield

The focus shield should then be cut to size to fill the area outside the width of the substrate.

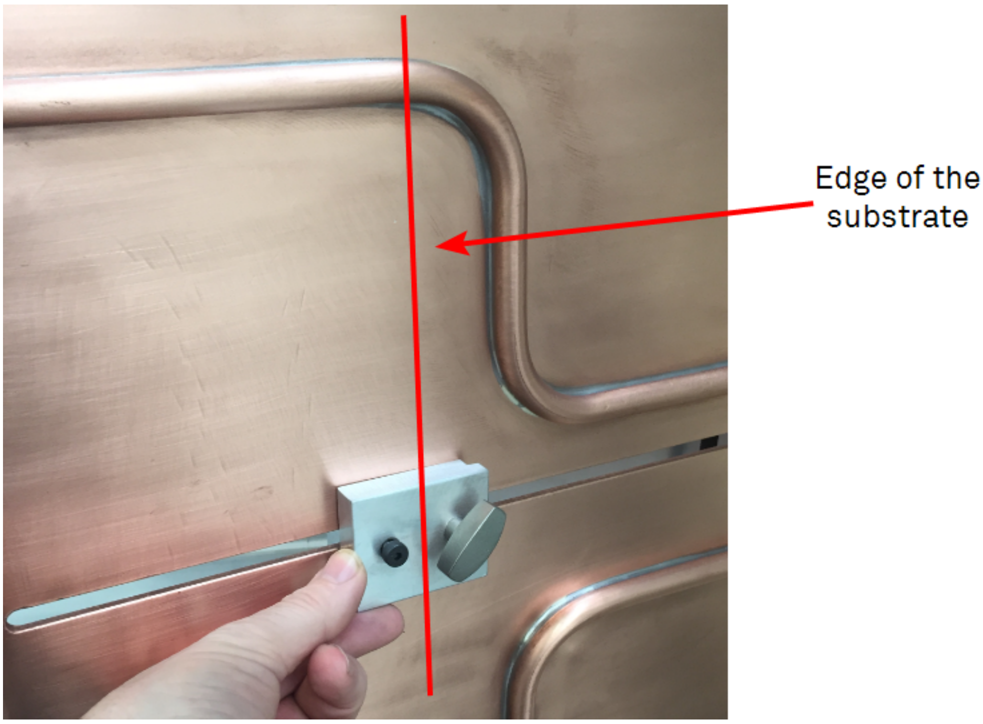

Figure 6.7-4 Photograph showing user moving the anti-back side treatment shield to align behind the substrate.

Figure 6.7-5 Photograph showing user clamping the focusing shield

Focussing Shields

The purpose of the aluminium “focussing shields” to stop a “high density” plasma effect being created outside the substrate width.

There are also called impedance shields.

A limited number of spare angle pieces are supplied with the machine.

The material is a standard aluminium angle with a size of 2x2x1/4 inch and the material code is BS1474 H30(6082)

	Note !
	**If the aluminium angle “focussing shields” are not installed, excess gas will be used in the process.** **Another negative effect is that the plasma intensity will be spread throughout the treater rather than just in the area where the substrate is located.**

6.8 Threading the Substrate through the Plasma Treater Unit

As the substrate to be coated has to pass through the plasma treater, there are a few basic steps to take in carrying out this procedure.

1. Check the width of the substrate. If wider than the previous cycle, release the anti-backside treatment shields and move them out to their extreme location.
2. Use a plastic guide to attach a threader or the substrate to and pull the threader/ substrate through the treater.
3. Continue to thread the material through the machine and apply tension to the substrate.
4. Inch the substrate through the machine to ensure that it is central.
5. Set the anti-backside shields to the correct position for the substrate width.

	**Warning !**
	Do not attempt to make any changes to the positions of any shielding if tension is applied to the substrate. **Turn off the movement isolator and LOCK it out, before entering this area.**

Figure 6.8-1 Anti-back side treatment shield

	**Note !**
	From experience, it has been found that end users manufacture sets of focussing shields in pairs to their common width substrates that allows the positioning to be carried out, prior to the substrate being placed in the machine.

Problems that can occur

- Anti-backside shields set too far from the substrate edge:- Can cause web breaks, or scratching
- Adhesive tape or other debris left inside the treater assembly:- Can cause high level arcing or web breaks

6.9 Cleaning and “Burn in” requirement

This chapter describes the cleaning and burn in requirements for the plasma treater system.

Planar Pre-treaters

From experience, plasma treaters require thorough cleaning initially to allow for plasma process stability.

The reason for this is that the electrodes form an oxide layer, which needs to be burned off by the formation of plasma on the surface.

The condition of the electrode surfaces also have an effect on plasma stability. For example, if there are many scratches on the surface or small gouges, these act to make the surface area of the oxide even worse and therefore more difficult to remove.

The condition of the electrode should be checked on a regular basis

Experience of this area shows damage has been seen on the electrode caused by , scratches, pitting and steel swarf.

Figure 6.9-1 Example of electrode surface damage.

This photograph shows some pitting found on the electrode surface.

Again, this type of damage needs to be removed using a fine grade of abrasive paper.

Figure 6.9-2 Electrode surface after cleaning.

Cleaning of the electrode plates

Cleaning should take place, depending upon process conditions. See chapter Section 8.0 Machine Cleaning Processes on page 348 for details.

Burn In Procedure

Following this extensive cleaning, it is recommended that the following procedure takes place.

1. Remove all substrate from the machine.
2. Close the machine and evacuate it as normally until the pressure in the winding zone is <3e^{-2}mbar
3. Select the DIAGNOSTIC PLASMA TEST function for the plasma treater.

Figure 6.9-3 DIAGNOSTICS - PLASMA HMI screen.

4. Apply process gas to bring the operation pressure to 6e^{-2} mbar
5. Power can now be applied to the plasma treater, using the enable function
   It is advisable to run the unit with an initial power set point of 2KW, leaving it for several minutes until the plasma stability is reached.
6. The power can then be incremented in 1KW (or 0.5KW) steps, waiting several minutes in between each increment to allow several rotations to remove the oxide layer.

6.10 High Oxygen Concentration Process Gas Safety

In machines that are operating at oxygen gas levels higher than 20% by volume of the total process gas environment, certain guidelines have to be adhered to.

Figure 6.10-1 Air dilution system on Busch Cobra pump

Extra safety equipment has been installed to ensure that the exhausted oxygen concentrated air, cannot exceed 23% of the surrounding atmosphere.

Oxygen supply lines and safety

There are specific guidelines relating to oxygen gas lines, and regulators. Bobst Manchester Ltd follow good industry practices in the design and installation of these lines. The customer is advised to follow the same good practices to reduce the risk of an incident.

Preparation

- Pre-cleaning of parts should be performed before final cleaning for oxygen service to remove any gross contamination such as large quantities of oils, greases, and inorganic particulates.
- All cleaning & pre-assembly must be done on a properly cleaned stainless steel work surface as far away from sources of contamination as possible
- All assembly tools and assembly surfaces should be cleaned with 99% Isopropyl alcohol prior to use.
- Care should be taken not to touch part surfaces that will be in direct oxygen service except with clean, powder-free, nitrile gloves.

Cleaning Procedure

- Parts are to be cleaned/degreased using a proprietary cleaning agent 99% Isopropyl alcohol.
- Take all necessary precautions when handling these chemicals Read the MSDS sheet.
- After degreasing dry parts completely by purging with dry nitrogen gas.

Warning !

Factory/system compressed air should not be used for drying parts.

Visually inspect parts for the presence of contaminants under strong white light and for the absence of accumulations of lint fibres. Any visual contaminate is cause for re-cleaning.

Cleaning Materials

Solvents/Detergents: - 99% Isopropyl alcohol.

Cleaning cloths should be clean, lint free & free from oil or grease. Cleaning cloths should be removed after use.

Protective Equipment:-

- Disposable non-powdered nitrile or latex gloves to be worn when handling cleaned parts.
- Suitable PPE to be worn (determined by cleaning agent).

Assembly Procedure

All assembly tools and assembly surfaces should be cleaned with 99% Isopropyl alcohol prior to use. Care should be taken not to touch part surfaces that will be in direct oxygen service except with clean, powder-free, Nitrile gloves (Latex gloves are organic and therefore should not be used when assembling oxygen equipment).

Warning !

Standard PTFE tape contains oil-based hydrocarbons & must never be used for thread sealing anywhere on the oxygen supply line.

Only thread sealing products suitable for oxygen use should be used (Cobas Green Oxygen Thread Sealing Tape – BML Part No 268166G).

Parts should be assembled immediately after cleaning.

Pre-assembly must be done on a properly cleaned stainless steel work surface as far away from sources of contamination as possible.

- Assemble all parts onto the machine.
- Care must be taken to avoid contamination during assembly.
- Any item, which becomes contaminated during assembly, must be re-cleaned before installation.
- When assembly is complete, purge each Oxygen system with dry nitrogen gas.
- Leak test each system with dry nitrogen gas.
- A suitable oxygen rated leak test solution is required for the leak checking e.g. “Snoop” by Swagelok.

Warning !

Do not use a soap based leak test solution as it will contain fats and oils.

Gas Line Identification

Note !

All gas lines to be clearly labelled at connection points and at 2 metre intervals along the pipeline.

Labels must identify the specific gas type (oxygen, nitrogen, argon, etc.)

Further Reference Document

- EIGA – IGC Doc. 33/06/E – Cleaning of equipment for oxygen services

SECTION 7.0 MACHINE OPERATIONAL SEQUENCES

This section provides a series of flowcharts that describe the machine operational sequences:

7.1 Start-up Sequence of the Machine

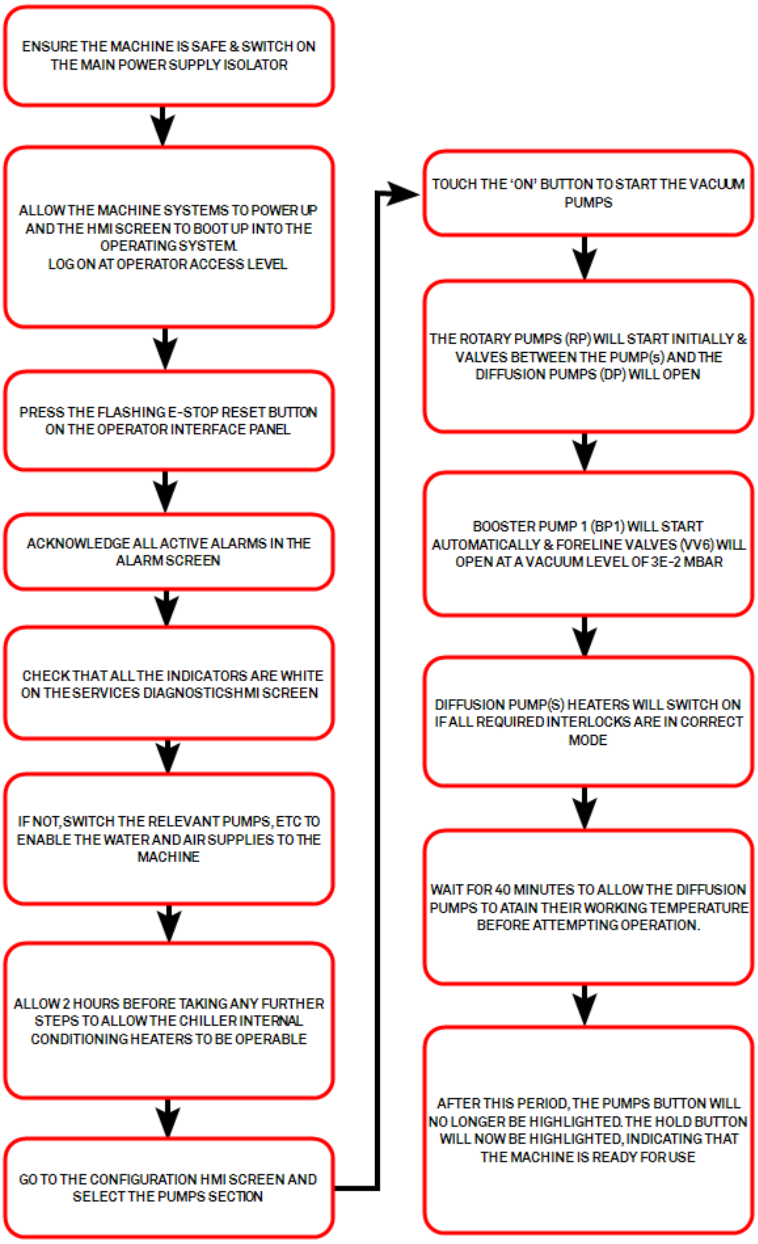

Figure 7.1-1 Start-up Sequence

7.2 Vacuum System Sequences

This chapter describes the vacuum system sequences in a series of flowcharts.

Vacuum “Hold” Condition

The “HOLD” condition is a description given to the vacuum pumping system when the vacuum levels are held constant within the chamber. This pressure may be at atmosphere or at the end of a cycle.

The “HOLD” condition is used as a buffer condition between changing from “VACUUM” to “VENT” for example.

The hold mode is indicated by a triangle on the “HOLD” button on the vacuum HMI screen.

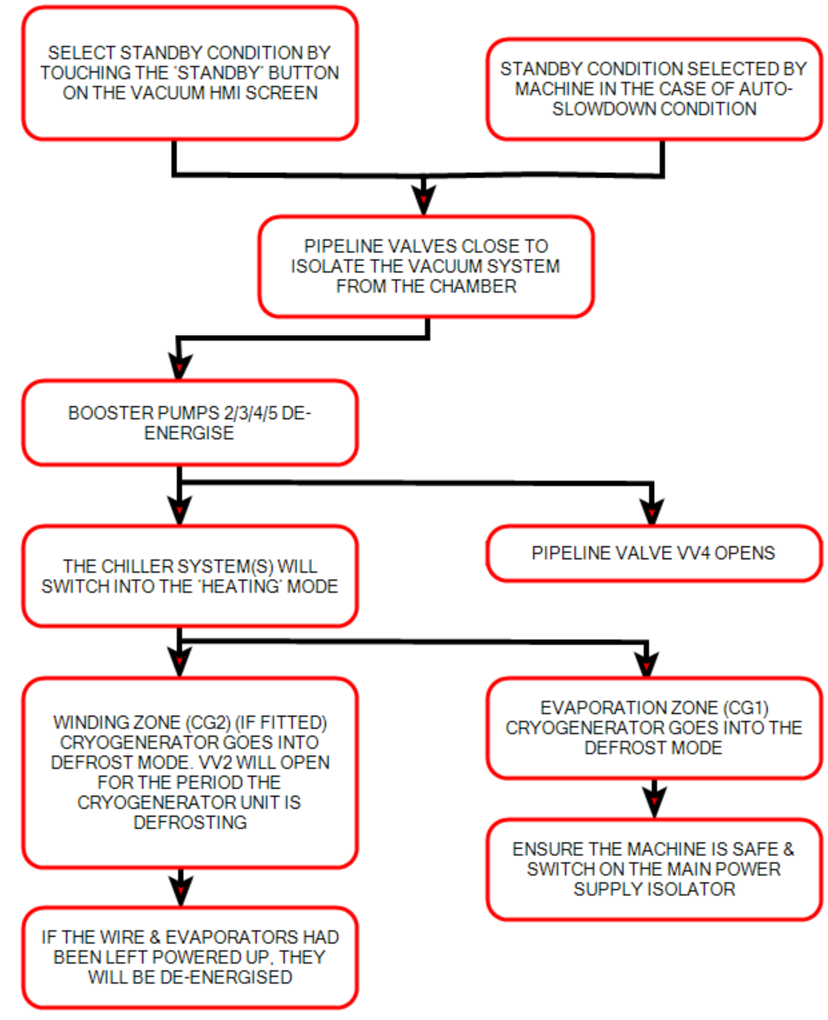

Figure 7.2-1 Standby Sequence

The pumping system is designed to evacuate the chamber in an efficient and safe manner from atmosphere to the working pressure, once the chamber has been closed.

Warning !

No adjustments must be made to the sequence without contacting and gaining the permission in writing from Bobst Manchester Ltd.

The flow diagram for this sequence can be seen below.

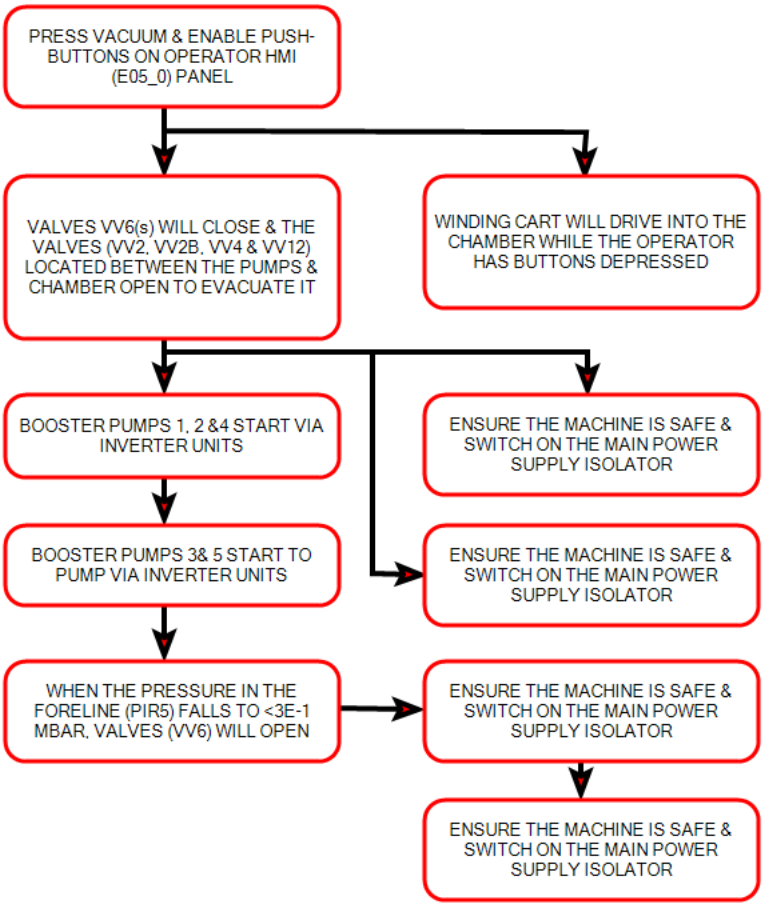

Figure 7.2-2 Machine Pumpdown Sequence

Chamber Vent Sequence

Before opening the chamber at the end of a process cycle, the chamber has to be vented to atmosphere.

The vacuum pumping system will be isolated from the chamber and the air surrounding the machine will be vented into the chamber. The following block diagram explains this sequence;-